Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION 2024

Name of the Course: M.Tech

Semester: 3rd

Name of the Paper: Soft Computing

Paper Code: MCS-301

Time: 3 Hours

Maximum Marks: 100

Note: -

(i) All questions are compulsory.

(ii) Answer any two sub-questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

Q1	(20marks)	**CO1**
(a)	Explain the differences between biological neural networks and artificial neural networks.	
(b)	With the help of a diagram, describe the working of a McCulloch-Pitts neuron model. What are its limitations?	
(c)	Compare feed-forward and feedback network. In what ways is bipolar representation better than binary representation	
Q2	(20 marks)	**CO2**
(a)	What are the activation functions in ANN? Explain the importance of different activation functions with examples.	
(b)	What are the various types of fuzzy composition techniques?	
(c)	Implement OR function with binary inputs and bipolar targets using perceptron training algorithm?	
Q3	(20 marks)	**CO3**
(a)	Compare and contrast the Single Layer Perceptron and Multi-Layer Perceptron.	
(b)	Consider two fuzzy sets $A=\{\frac{1}{2.0}+\frac{0.65}{4.0}+\frac{0.5}{6.0}+\frac{0.35}{8.0}+\frac{0}{10.0}\}$ B= $A=\{\frac{0.2}{2.0}+\frac{0.55}{4.0}+\frac{0.7}{6.0}+\frac{0.65}{8.0}+\frac{1}{10.0}\}$ Find the following (1) AUB (2) A∩B (3) AU$\bar{A}$ (4) $\overline{AUB}$ (5) $\overline{A \cap B}$	
(c)	Implement XOR function using McCulloch-Pitts neuron (consider binary data)	
Q4	(20 marks)	**CO4**
(a)	Define defuzzification. Discuss in detail on the various types of defuzzification methods.	
(b)	Define fuzzy logic and crisp logic. With suitable examples, explain the operations and properties of fuzzy sets, crisp set, fuzzy relations, and crisp	

	relations.	
(c)	Briefly explain the basic concepts of Genetic algorithms. Write short notes on 1. Fitness function 2. Mutation 3. Crossover	
Q5	(20 marks)	**CO5**
(a)	Using the genetic algorithm process, minimize the function $f(x)=x^2+5x$ Assume the necessary operators for the process of your own.	
(b)	Illustrate the role of membership functions in fuzzy systems. Explain different types of membership functions.	
(c)	Discuss the concept of Self-Organizing Maps (SOM), focusing on its neural network architecture. Discuss how SOM works and highlight its key application areas with examples.	